

Cloud Computing

Lab-1

Goal : The goal of this lab is to launch a single EC2 instance in a public subnet accessible over the Internet via SSH.

Architecture Diagram:

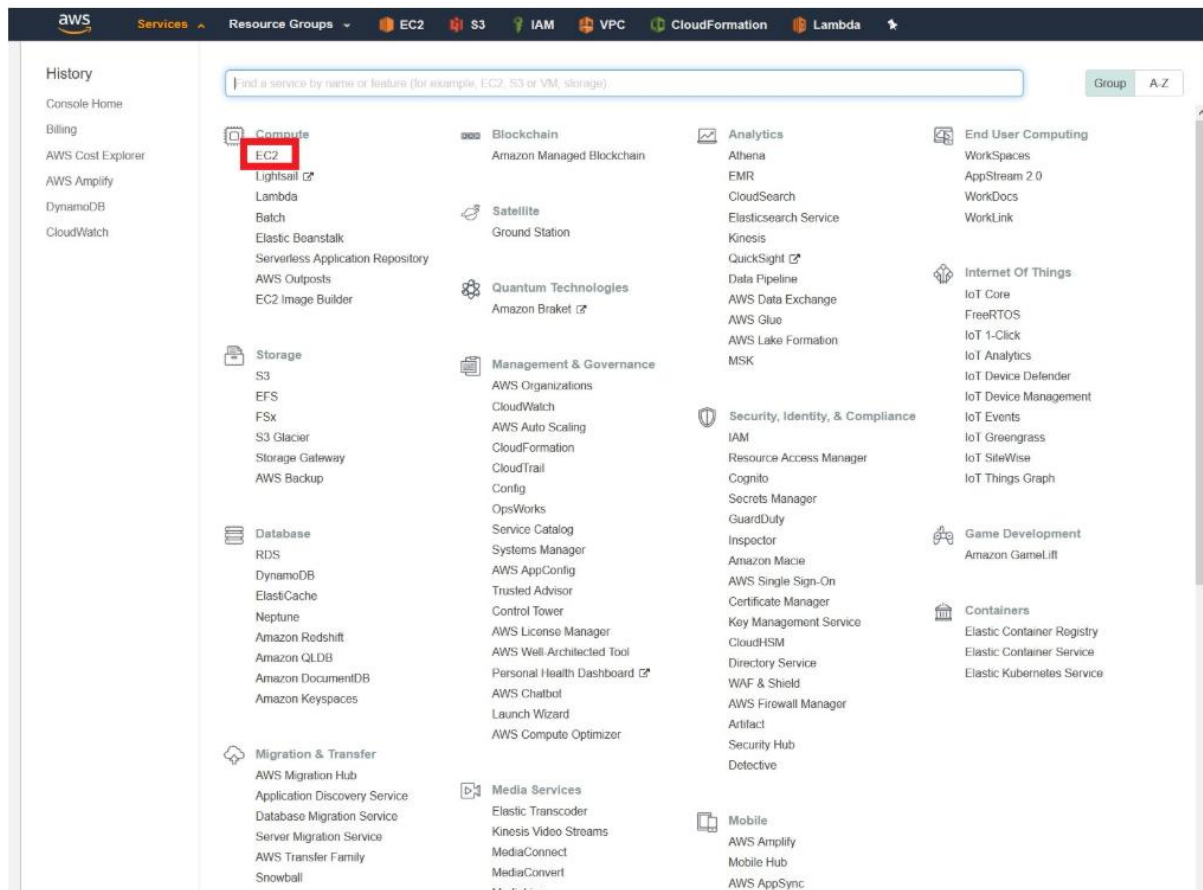


Overview

In order to achieve the goal of this lab, you will have to go through the following steps:

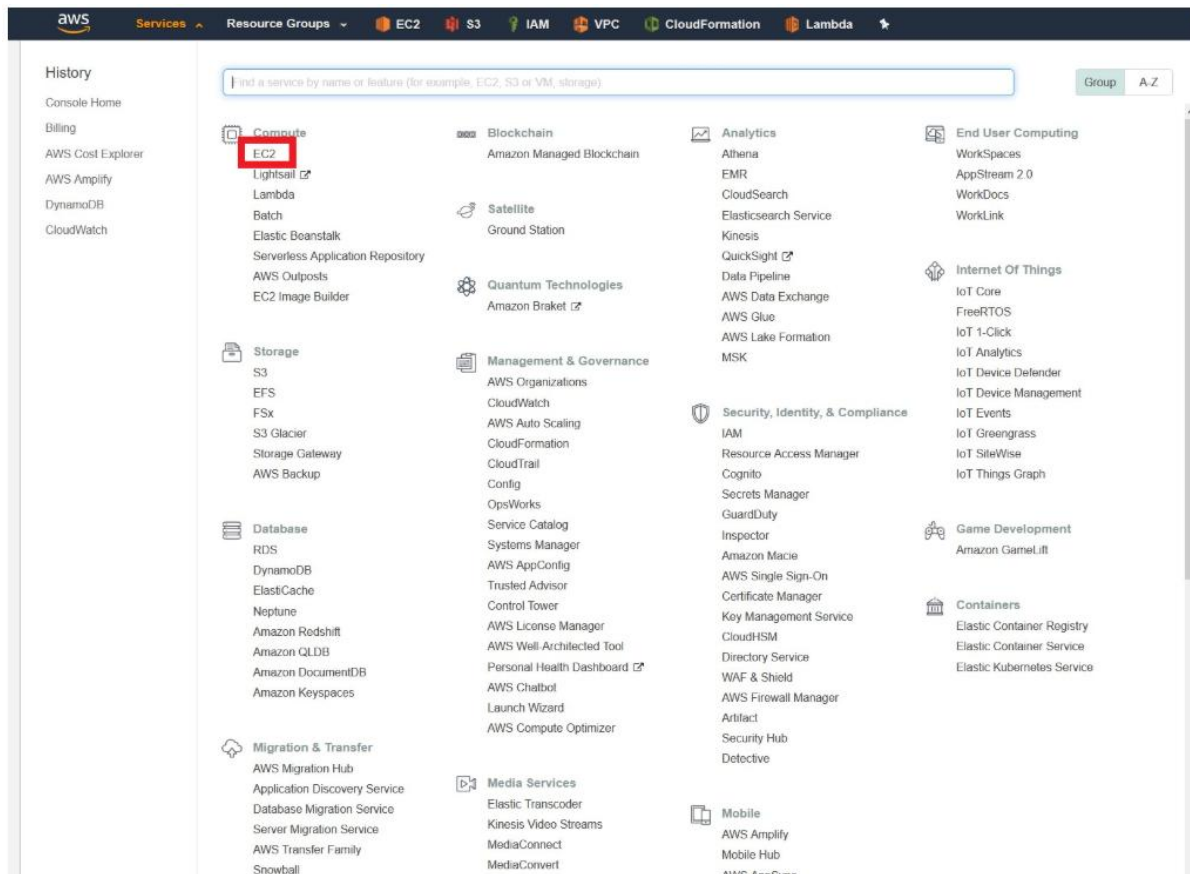
1. Choose the operating system by selecting the [Amazon Machine Image \(AMI\)](#).
2. Define the virtual hardware configuration by choosing an [Instance Type](#).
3. Review the network settings.
4. Review the storage settings.
5. Create tags (optional).
6. Configure the [Security Group](#) rules (firewall).
7. Launch the instance (choosing or creating an [EC2 key pair](#)).

Go to [AWS Console](#) and login with your credentials. Click on Services at the top left, then choose EC2 in the Compute section.



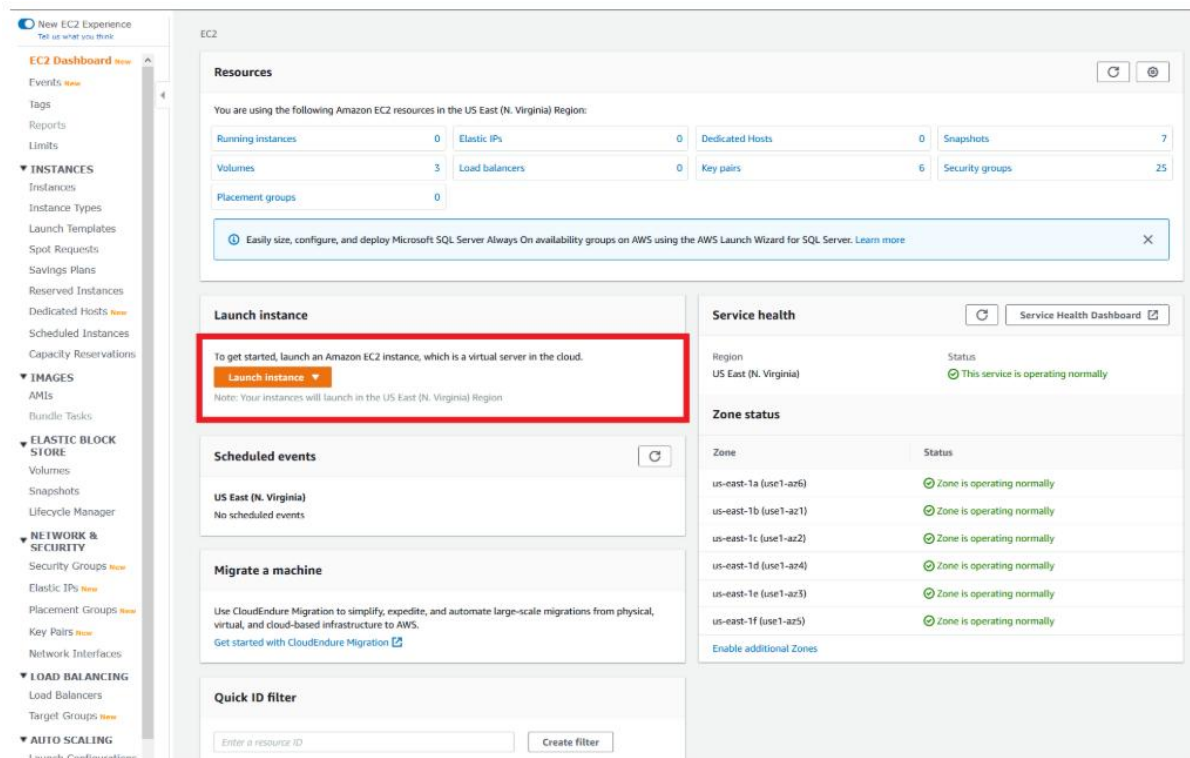
Step 1 - Choose the AMI

In the EC2 service menu, click on *Launch instance* and select *Launch Instance*.



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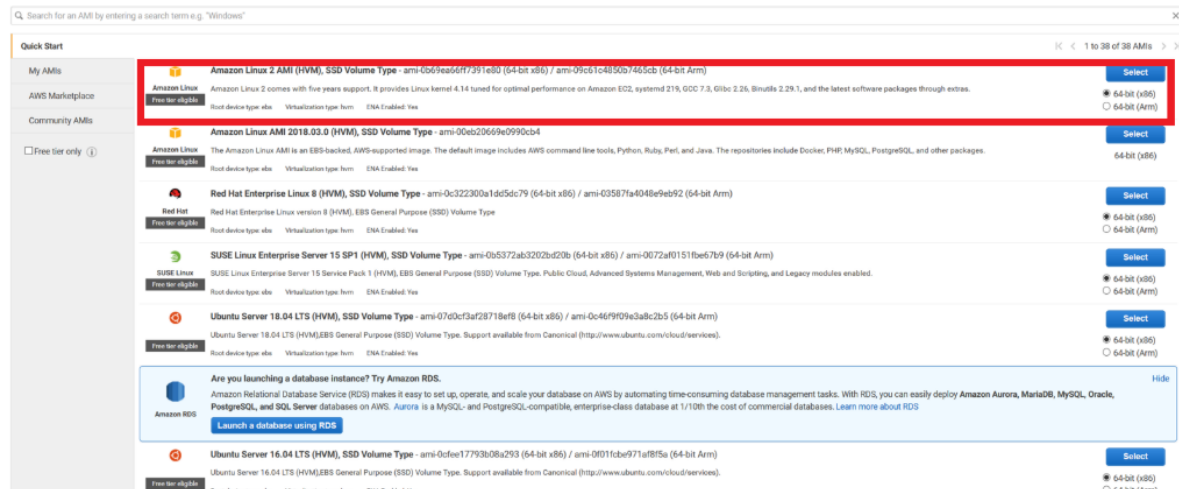


Select *Amazon Linux 2 AMI*. An AMI is a template that contains the software configuration (operating

system, application server, and applications) required to launch your instance. In this first lab, we will stick to the x86 architecture.

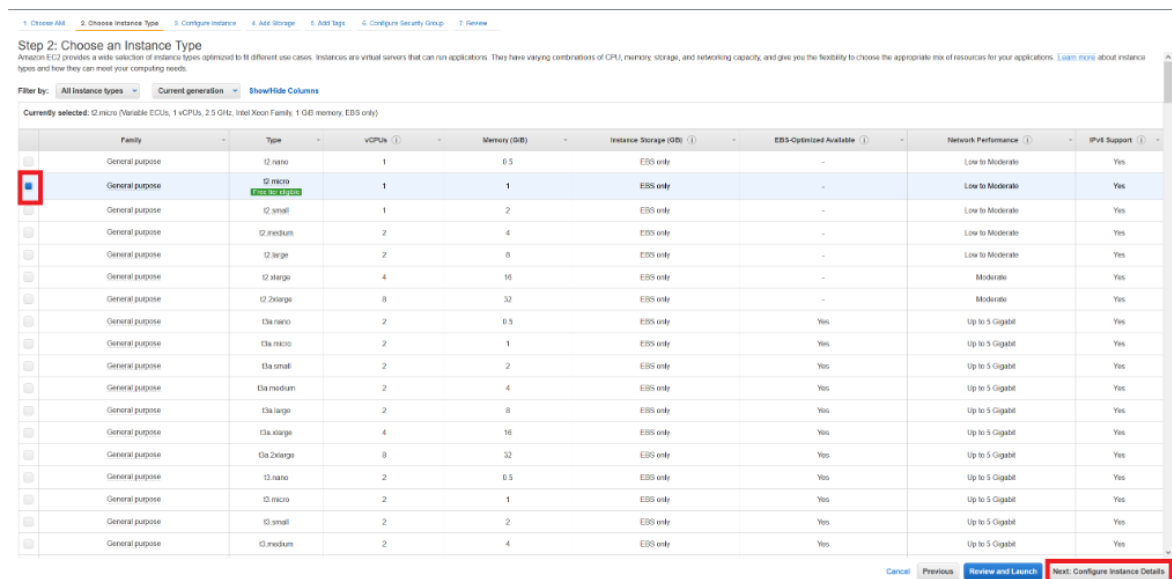
Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.



Step 2 - Define the Instance Type

The instance type defines the CPU and memory capacity. It also specifies the storage architecture that the instance will support, along with network performance available. You can use the suggested free-tier eligible instance type, and then go on to *Next: Configure Instance Details*.



Step 3 - Network Settings

Make sure you select the public default subnet and that the *Auto-assign Public IP* setting is enabled for that subnet. You may leave all the other settings with their default values.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

Number of instances: 1 [Launch into Auto Scaling Group](#)

Purchasing option: ☐ Request Spot instances

Network: vpc-2223a659 (default (default)) [Create new VPC](#)

Subnet: No preference (default subnet in any Availability Zone) [Create new subnet](#)

Auto-assign Public IP: **Use subnet setting (Enable)**

Placement group: ☐ Add instance to placement group

Capacity Reservation: Open [Create new Capacity Reservation](#)

IAM role: None [Create new IAM role](#)

Shutdown behavior: Stop

Stop - Hibernate behavior: ☐ Enable hibernation as an additional stop behavior

Enable termination protection: ☐ Protect against accidental termination

Monitoring: ☐ Enable CloudWatch detailed monitoring [Additional charges apply](#)

Tenancy: ☐ Shared - Run a shared hardware instance [Additional charges may apply when launching Dedicated instances](#)

Elastic Inference: ☐ Add an elastic inference accelerator [Additional charges apply](#)

T2/T3 Unlimited: ☐ Enable [Additional charges may apply](#)

File systems: [Add file system](#) [Create new file system](#)

Advanced Details

Metadata accessible: Enabled

Metadata version: V1 and V2 (token optional)

Metadata token response hop limit: 1

User data: ☒ As text ☐ As file ☐ Input is already base64 encoded

[Cancel](#) [Previous](#) [Review and Launch](#) **[Next: Add Storage](#)**

Go on to *Next: Add Storage*.

Step 4 - Storage Settings

You can accept the default settings which creates an EBS root volume.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type	Device	Snapshot	Size (GiB)	Volume Type	IOPS	Throughput (MiB/s)	Delete on Termination	Encryption
Root	/dev/nvda	snap-074e8f0568ee430f1	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypted

[Add New Volume](#)

Free tier eligible customers can get up to 30 GiB of EBS General Purpose (SSD) or Magnetic storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

[Cancel](#) [Previous](#) [Review and Launch](#) **[Next: Add Tags](#)**

Go on to *Next: Add Tags*.

Step 5 - Tags

Tags is a way to add textual metadata to AWS Resources to help you manage your cloud environment. For now, you can skip this step.

1 Choose AMI 2 Choose Instance Type 3 Configure Instance 4 Add Storage 5 Add Tags 6 Configure Security Group 7 Review

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.
A copy of a tag can be applied to volumes, instances or both.
Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key (128 characters maximum)	Value (256 characters maximum)	Instances (1)	Volumes (1)
This resource currently has no tags.			

Choose the **Add tag** button or click to add a Name tag.
Make sure your [IAM policy](#) includes permissions to create tags.

Add Tag (Up to 50 tags maximum)

Cancel Previous **Review and Launch** Next: Configure Security Group

Go on to *Next: Configure Security Group*.

Step 6 - Security Group

The *Security Group* is a firewall for the instance. Create a new security group named *ssh-access* with a rule that allows SSH from anywhere (or choose *My IP* if you prefer).

1 Choose AMI 2 Choose Instance Type 3 Configure Instance 4 Add Storage 5 Add Tags 6 Configure Security Group 7 Review

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☒ Create a new security group ☐ Select an existing security group

Security group name: **ssh-access**

Description: **lab001 security group**

Type (1)	Protocol (1)	Port Range (1)	Source (1)	Description (1)
SSH	TCP	22	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop

Add Rule

Warning
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Previous **Review and Launch**

Go on to *Review and Launch*.

Step 7 - Launch Instance

You can review all the settings until this step in the next screen.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 7: Review Instance Launch

Please review your instance launch details. You can go back to edit charges for each section. Click **Launch** to assign a key pair to your instance and complete the launch process.

⚠ Improve your instances' security. Your security group, default, is open to the world.
 Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only. You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details [Edit AMI](#)

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-09d5fab7f8377f6c

Amazon Linux 2 comes with two years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.28, Binutils 2.29.1, and the latest software packages through extras.

Find out details about this AMI. Your device type: etc. Virtualization type: hvm

Instance Type [Edit instance type](#)

Instance Type	ECUs	vCPUs	Memory (GiB)	Instance Storage (GiB)	EBS-Optimized Available	Network Performance
t2.micro	Variable	1	1	EBS only	-	Low to Moderate

Security Groups [Edit security groups](#)

Security Group ID	Name	Description
sg-b53796c3	default	default VPC security group

All selected security groups inbound rules

Type	Protocol	Port Range	Source	Description
All traffic	All	All	sg-b53796c3 (default)	
SSH	TCP	22	0.0.0.0/0	
SSH	TCP	22	:::0	

[Instance Details](#) [Edit instance details](#)

[Storage](#) [Edit storage](#)

[Cancel](#) [Previous](#) [Launch](#)

If everything is Ok, go on to *Launch*. In the new windows, for the key-pair you can create a new key pair (name it lab-001). Download the key pair to your computer

Select an existing key pair or create a new key pair

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair

Key pair name

lab-001

Download Key Pair

You have to download the **private key file** (*.pem file) before you can continue. **Store it in a secure and accessible location.** You will not be able to download the file again after it's created.

[Cancel](#) [Launch Instances](#)

You will be forwarded to a new screen with the launch status.

Launch Status

 **Your instances are now launching**
The following instance launches have been initiated: i-0437b3dc76388b7f9 [View launch log](#)

 **Get notified of estimated charges**
Create billing alerts to get an email notification when estimated charges on your AWS bill exceed an amount you define (for example, if you exceed the free usage tier).

How to connect to your instances

Your instances are launching, and it may take a few minutes until they are in the **running** state, when they will be ready for you to use. Usage hours on your new instances will start immediately and continue to accrue until you stop or terminate your instances. Click **View instances** to monitor your instances' status. Once your instances are in the **running** state, you can **connect** to them from the Instances screen. [Find out](#) how to connect to your instances.

▼ Here are some helpful resources to get you started

- [How to connect to your Linux instance](#)
- [Learn about AWS Free Usage Tier](#)
- [Amazon EC2: User Guide](#)
- [Amazon EC2: Discussion Forum](#)

While your instances are launching you can also

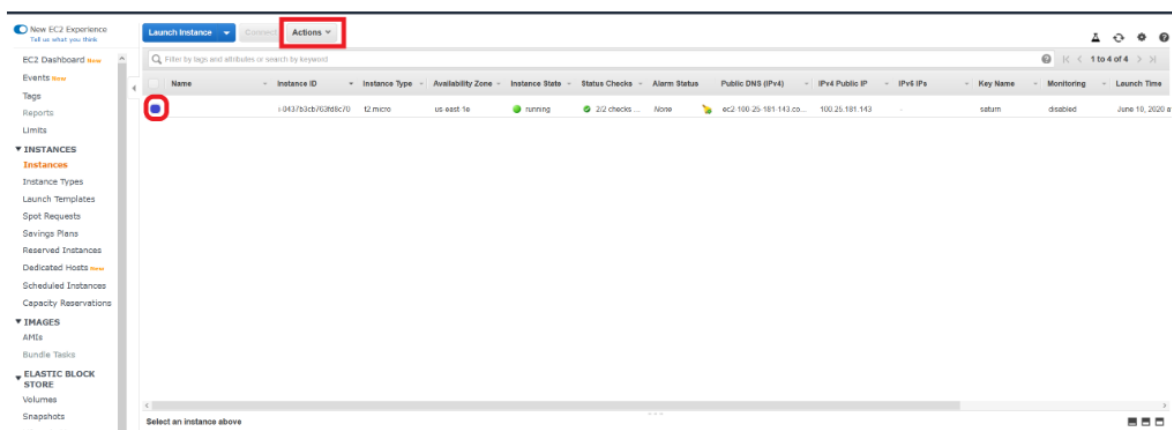
- [Create status check alarms](#) to be notified when these instances fail status checks. (Additional charges may apply)
- [Create and attach additional EBS volumes](#) (Additional charges may apply)
- [Manage security groups](#)

[View instances](#)

Go on to *View Instances* to return to the EC2 Console. There you will a list of existing instances and the new instance you just created.

Test & Validation

Once your instance is up and running, select it on the EC2 console and click on *Connect*.



Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (IPv4)	IPv4 Public IP	IPv4 IPs	Key Name	Monitoring	Launch Time
	i-0437b3dc76388b7f9	t2.micro	us-east-1a	running	2/2 checks ...	None	ec2-100-25-181-143.co...	100.25.181.143	-	saturn	disabled	June 10, 2020 a

From the new window, copy the example ssh command-line. It will have the following format:

```
$ ssh -i "lab-001.pem" ec2-user@ec2-100-25-181-143.compute-1.amazonaws.com
```

The *ec2-user* is the default user name in Amazon Linux AMIs. The hostname *ec2-100-25-181-143.compute-1.amazonaws.com* is the DNS name for your instance that AWS automatically defines. Open a terminal window and move to the folder where you saved the downloaded key file *lab-001.pem*. Change the permissions of the key file:

```
$ chmod 400 lab-001.pem
```

Paste and run the ssh command. You should be able to log into your instance. If you are a Windows user, you can also use Putty to connect. Follow this [link](#) for instructions.